

Pablo López Serna

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Education

Technical University of Madrid (UPM), Spain

Sep 2026 – Expected Jun 2027

Master in Financial Engineering and Innovation

- Advanced curriculum: Game Theory, Advanced Probability, Financial Mathematics, Machine Learning. Core modules led by industry practitioners and quantitative finance experts.

University of Salamanca, Spain

Sep 2022 – Jul 2026

Bachelor of Science in Physics – 95th Percentile of cohort

- Honours in Computational Physics and Statistical Physics. Coursework: Numerical Methods, Differential Equations, Statistical Physics, Quantum Mechanics.

Imperial College London

Jun 2025 – Jul 2025

Summer School in Machine Learning, Statistics, and Trading Systems

- Selective programme integrating machine learning, stochastic processes, and quantitative modeling for financial markets.

Skills

Programming & Tools: Python (NumPy, Pandas, scikit-learn, TensorFlow/Keras), Fortran, C++, SQL, Git, AWS, Docker.

Quantitative Methods: Probability Theory, Stochastic Calculus, Monte Carlo Methods, Optimization, Time Series Analysis, Game Theory.

Languages: Spanish (Native), English (Full Professional), French (Elementary), Italian (Elementary).

Projects & Research

Options Pricer – Monte Carlo & Black–Scholes Framework

[code]

Independent Project

- Built a Python pricing engine cross-validating Black-Scholes, Merton jump-diffusion, binomial tree, finite-difference, and Longstaff-Schwartz Monte Carlo methods with analytic Greeks; binomial and finite-difference prices converge to within 0.1% of the closed-form Black-Scholes benchmark.

Order Book Predictor – Neural Order Flow Forecasting

[code]

Independent Project

- Trained a neural network on limit-order-book snapshots to predict short-term mid-price direction, reaching 77.3% out-of-sample accuracy versus a 71.7% logistic-regression baseline.

Monte Carlo Simulation Algorithms for the Ising Model

[code]

Tutored project for Physics Degree

- Implemented Metropolis, Glauber, and Wolff Monte Carlo algorithms in Fortran; estimated the critical temperature within 0.15% of the exact Onsager solution via finite-size scaling.

Modeling of Topological Charge of Magnetic Structures Under Realistic Excitations

In progress

Tutored project for RSEF - GEFES Research Award

Awards

- RSEF - GEFES Research Awards for Undergraduate Students (2026)
- University of Salamanca Magna Cum Laude Award for Academic and Professional Excellence (2025)

Professional Experience

Interim Quant Analyst – Coral Homes / Lone Star Funds

Jul 2025 – Sep 2025

Madrid, Spain

- Conducted quantitative analysis on a REO land portfolio using Python and Pandas; built statistical models and dashboards to optimize disposition strategy and forecast portfolio risk.

Freelance Software Developer

Feb 2022 – Present

Independent Contractor

- Built automation, data-analytics pipelines, and backend infrastructure for clients in tech and finance; improved data processing throughput by 40%+ using Python, SQL, and AWS.

Lead Software Engineer – ViralPlan

Feb 2023 – Jan 2024

Barcelona, Spain

- Led a remote engineering team, reducing API latency by 35%, and designed internal dashboards for data-driven decisions.

Core Developer – Smart Crypto

Oct 2022 – May 2023

Barcelona, Spain

- Developed blockchain-integrated financial simulation tools and researched tokenomics for DeFi pricing mechanisms.

Interests

Pyfano volunteer/mentor (childhood cancer) | Profitable poker (3-max Hyper-Turbo) | Chess | Darts, horse riding & breeding